

.Active Device: cuda

Dataset root:

/kaggle/input/datasets/varun2ks05/rice-leaf-aug/Rice\_Leaf\_AUG

--- Training deit\_small\_patch16\_224 ---

Epoch [01/25] | Train Loss: 0.6873 Acc: 73.91% | Val Loss: 0.3421 Acc: 88.12%

--> Saved new best model (88.12%)

Epoch [02/25] | Train Loss: 0.2603 Acc: 91.15% | Val Loss: 0.1869 Acc: 94.52%

--> Saved new best model (94.52%)

Epoch [03/25] | Train Loss: 0.1258 Acc: 96.18% | Val Loss: 0.1651 Acc: 94.52%

Epoch [04/25] | Train Loss: 0.1182 Acc: 96.15% | Val Loss: 0.1309 Acc: 96.08%

--> Saved new best model (96.08%)

Epoch [05/25] | Train Loss: 0.0792 Acc: 97.22% | Val Loss: 0.1126 Acc: 96.08%

Epoch [06/25] | Train Loss: 0.0467 Acc: 98.63% | Val Loss: 0.1717 Acc: 94.78%

Epoch [07/25] | Train Loss: 0.0615 Acc: 98.17% | Val Loss: 0.1501 Acc: 96.34%

--> Saved new best model (96.34%)

Epoch [08/25] | Train Loss: 0.0455 Acc: 98.66% | Val Loss: 0.1329 Acc: 96.61%

--> Saved new best model (96.61%)

Epoch [09/25] | Train Loss: 0.0301 Acc: 99.09% | Val Loss: 0.0848 Acc: 98.30%

--> Saved new best model (98.30%)

Epoch [10/25] | Train Loss: 0.0289 Acc: 99.22% | Val Loss: 0.0665 Acc: 98.04%

Epoch [11/25] | Train Loss: 0.0072 Acc: 99.87% | Val Loss: 0.0821 Acc: 97.65%

Epoch [12/25] | Train Loss: 0.0242 Acc: 99.22% | Val Loss: 0.0760 Acc: 97.78%

Epoch [13/25] | Train Loss: 0.0133 Acc: 99.58% | Val Loss: 0.0743 Acc: 98.04%

Epoch [14/25] | Train Loss: 0.0161 Acc: 99.41% | Val Loss: 0.0713 Acc: 98.30%

Epoch [15/25] | Train Loss: 0.0082 Acc: 99.77% | Val Loss: 0.0679 Acc: 98.30%

Epoch [16/25] | Train Loss: 0.0031 Acc: 99.93% | Val Loss: 0.0654 Acc: 98.43%

--> Saved new best model (98.43%)

Epoch [17/25] | Train Loss: 0.0027 Acc: 99.87% | Val Loss: 0.0684 Acc: 98.30%

Epoch [18/25] | Train Loss: 0.0017 Acc: 100.00% | Val Loss: 0.0688 Acc: 98.43%

Epoch [19/25] | Train Loss: 0.0039 Acc: 99.90% | Val Loss: 0.0593 Acc: 98.56%

--> Saved new best model (98.56%)

Epoch [20/25] | Train Loss: 0.0015 Acc: 100.00% | Val Loss: 0.0621 Acc: 98.56%

Epoch [21/25] | Train Loss: 0.0019 Acc: 99.97% | Val Loss: 0.0586 Acc: 98.69%

--> Saved new best model (98.69%)

Epoch [22/25] | Train Loss: 0.0012 Acc: 100.00% | Val Loss: 0.0603 Acc: 98.69%

Epoch [23/25] | Train Loss: 0.0014 Acc: 99.97% | Val Loss: 0.0596 Acc: 98.56%

Epoch [24/25] | Train Loss: 0.0027 Acc: 99.93% | Val Loss: 0.0586 Acc: 98.69%

Epoch [25/25] | Train Loss: 0.0016 Acc: 99.97% | Val Loss: 0.0585 Acc: 98.69%

Completed in 16m 12s. Peak Val Accuracy: 98.69%

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FINAL EVALUATION METRICS

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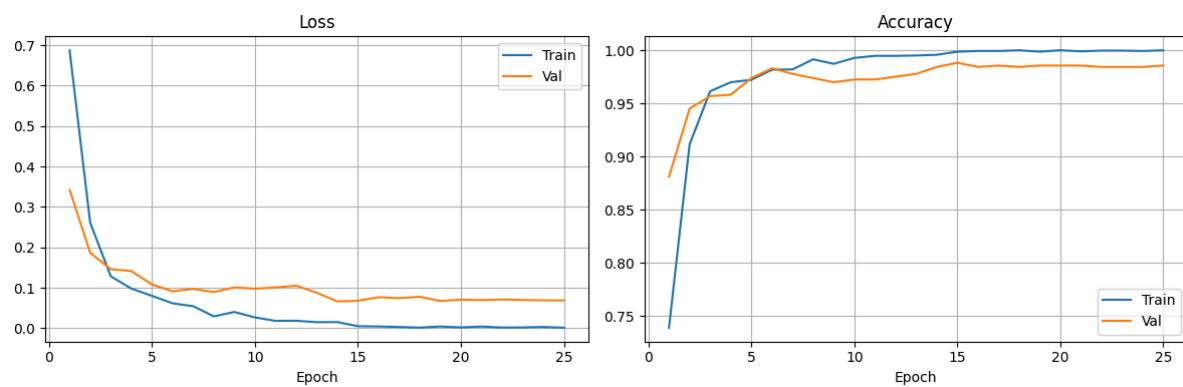
Overall Accuracy : 98.69%  
Macro Precision : 98.70%  
Macro Recall : 98.69%  
Macro F1-Score : 98.69%

--- Detailed Classification Report ---

|                       | precision | recall | f1-score | support |
|-----------------------|-----------|--------|----------|---------|
| Bacterial Leaf Blight | 0.9921    | 0.9843 | 0.9881   | 127     |
| Brown Spot            | 0.9922    | 0.9845 | 0.9883   | 129     |
| Healthy Rice Leaf     | 0.9924    | 1.0000 | 0.9962   | 131     |
| Leaf Blast            | 0.9695    | 1.0000 | 0.9845   | 127     |
| Leaf scald            | 0.9840    | 0.9762 | 0.9801   | 126     |
| Sheath Blight         | 0.9919    | 0.9762 | 0.9840   | 126     |
| accuracy              |           |        | 0.9869   | 766     |
| macro avg             | 0.9870    | 0.9869 | 0.9869   | 766     |
| weighted avg          | 0.9871    | 0.9869 | 0.9869   | 766     |

--- Class-wise Accuracy ---

Bacterial Leaf Blight : 98.43%  
    Brown Spot : 98.45%  
    Healthy Rice Leaf : 100.00%  
    Leaf Blast : 100.00%  
    Leaf scald : 97.62%  
    Sheath Blight : 97.62%



**DeiT Model Training Metrics**

